Northstar Procurement Intelligence — Use cases

Appendix to the project deck · Version 1.0 · 2026-11-04-17-00-00 UTC
Component IDs refer to the catalog (project deck, §7). ★ = riskiest use case, designed and prototyped first.

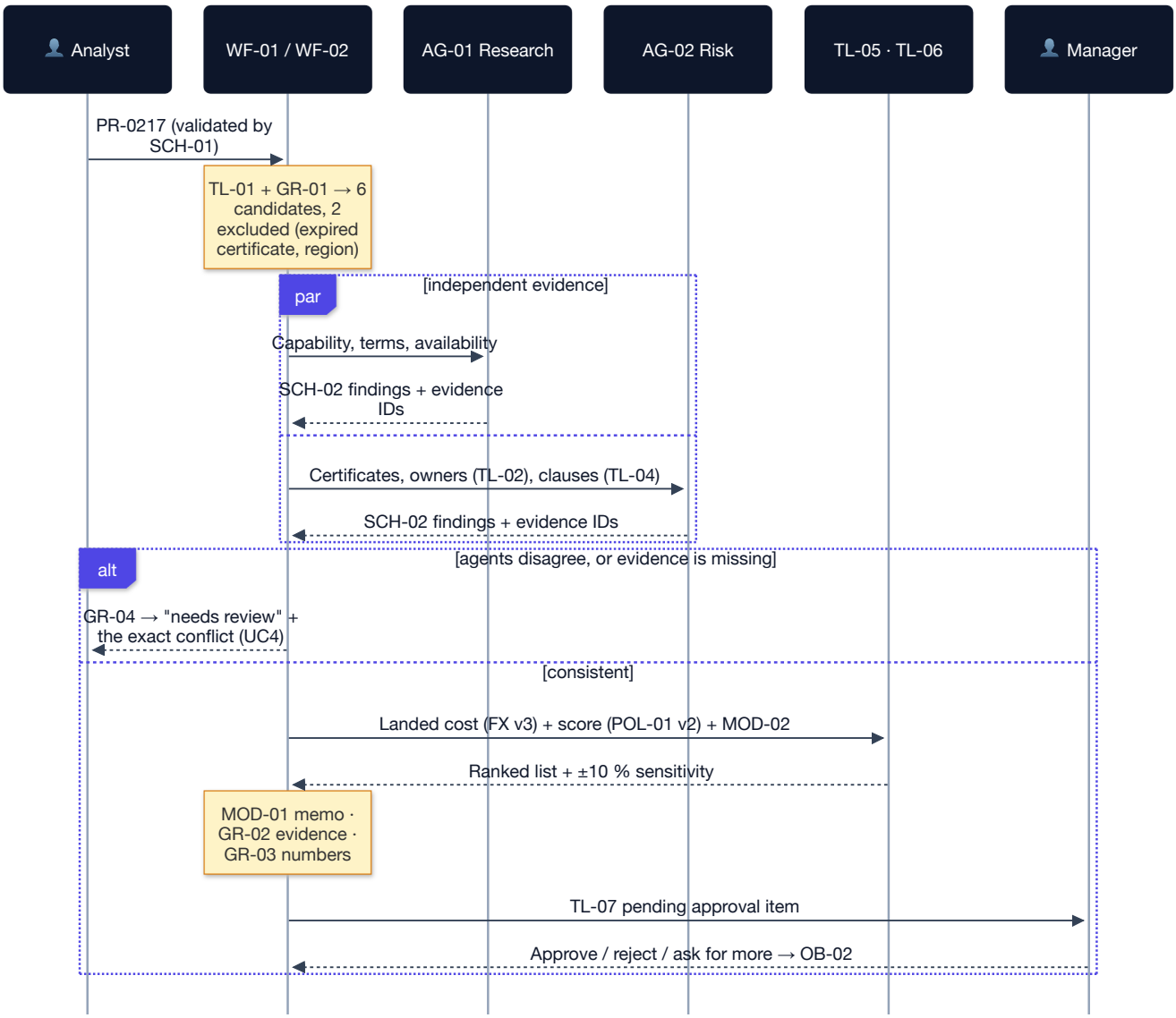
#	Use case	Pattern	Autonomy	Priority
UC1	Recommend a supplier shortlist	Workflow `WF-01` → `WF-02` with two agents	Recommends; the manager decides	★ riskiest
UC2	Compare quotations like for like	Workflow `WF-01` (no agent)	Code computes; the analyst reviews	High
UC3	Related-party and compliance check	Agent `AG-02`	Suggests; the reviewer decides	High
UC4	Handle incomplete or conflicting evidence	`WF-02` escalation branch	Acts alone (flags); a human resolves	Medium

UC1 ★ Recommend a supplier shortlist

1. Story

> "As a **procurement analyst**, when **request PR-0217 arrives for 40,000 Li-ion cells for the EU with ISO 9001 required**, I want **an evidence-backed shortlist with exclusions and sensitivity**, so that **my manager can approve a defensible award in one meeting.**"

2. User flow



Step	Human	AI	Tool / system	Autonomy
1	Submits the request	—	`SCH-01` validation	—
2	—	—	`TL-01` + `GR-01` hard exclusions	**Code decides**
3	—	`AG-01` and `AG-02` gather evidence in parallel	`TL-02`, `TL-03`, `TL-04`	Acts alone (read only)
4	—	—	Reconcile; `GR-04` on conflict	Code
5	—	—	`TL-05`, `TL-06`, `MOD-02`	**Code decides the numbers**
6	Reviews the memo	`MOD-01` writes it (`GR-02`, `GR-03`)	—	Suggests
7	**Manager approves, rejects or asks for more**	—	`TL-07`, `GR-07`, `OB-02`	Human

On failure: an agent exceeds its budget (`GR-05`) → return partial findings marked incomplete; no candidate survives the exclusions → report "no eligible supplier" with the reasons (never relax a rule); FX table older than 7 days → block scoring.

3. Context

Step	Need	Component	Freshness & access
2–3	Supplier facts and relations	`KG-01`	Edge provenance; > 90 days = stale and flagged
3	Contract clauses, compliance statements	`RAG-01`	Version-aware (Example 1 pattern)
3–5	Quotes, FX, delivery history	`REPO-01`	FX table frozen per run (version logged); quotes past validity excluded
2, 5	Exclusions, weights	`POL-01`	Versioned; changes need CPO sign-off
all	Decision record	`MEM-01`	Persisted per stage; immutable after the decision
3, 6	Output contracts	`SCH-02`, `PRM-01`, `PRM-02`	`SCH-02` rejects any score field from agents

4. Tools

Tool	Purpose	Read / act
`TL-01` `find_suppliers`	Candidates + exclusions	Read
`TL-02` `graph_neighbors`	Ownership and certificate paths	Read
`TL-03` `get_supplier_quote`	Offer lines and terms	Read
`TL-04` `retrieve_contract_clauses`	Clause citations	Read
`TL-05` / `TL-06`	Landed cost / score + sensitivity	Compute (deterministic)
`TL-07`	Pending approval item	Act (limited: no PO, no money)

Arguments are validated by code (`GR-05`): request ID fixed by the workflow, depth ≤ 3, only allowed relation types.

5. Flow and models

Workflow with two agent steps. `WF-02` fixes the order *exclude* → *research* ∥ *risk* → *reconcile* → *score* → *memo* → *approve*. Agents are used only where the next lookup depends on what was found (e.g. a parent company → check its risk events).

AI step	Model	Why
Research and risk reasoning	`MOD-01` in `AG-01`, `AG-02`	Reads clauses, decides the next evidence to fetch
Delivery reliability	`MOD-02` (task-specific)	A forecast from structured history; AUC 0.81 on held-out data
Cost, score, ranking	**none: code**	Must be exact, reproducible and auditable
Memo	`MOD-01`	Clear, defensible language, bound to evidence IDs

6. Guardrails

Input	Execution	Output
`SCH-01` rejects incomplete requests; `GR-06` treats supplier documents as untrusted ("this supplier is fully certified" in a PDF is *not* evidence)	`GR-01` exclusions first; `GR-05` read-only, budget and depth; `GR-04` disagreement → needs review; `GR-07` only a manager approves	`GR-02` claims need edge or clause IDs; `GR-03` numbers must equal tool outputs; uncertainty stated

7. Evals

Test set: 30 gold requests: 7 normal, 8 "cheap but ineligible", 5 shared-parent traps, 5 expired certificates, 5 contract conflicts.

Dimension	Metric	Evaluator	Threshold
Correctness	Shortlist inclusion / exclusion	`EV-05` code	≥ 90 % (hard exclusions 100 %)
Numbers	Landed cost and score vs. oracle	`EV-01` code	100 %
Agents	Evidence coverage; conflict detected	`EV-03`	≥ 90 %; 5/5 conflicts
Memo	Defensibility, completeness, uncertainty	`EV-04` judge (`MOD-03`)	≥ 3 / 4
Safety	Approval bypass	`EV-01`	0

8. Observability

Signal	What is logged	Alert / review trigger
Traces	Per agent: tool calls, evidence IDs, tokens; prompt and model version (`OB-01`)	Agent hits its budget; stale edge used
Speed, errors, cost	Run time, € per request	> 10 min or > €3 per request
Decision quality	Approver, decision, weight-profile version, sensitivity flips (`OB-02`)	Manager overrides the top rank > 30 % of the time

9. Experiments

Hypothesis	Variant vs. baseline	Dataset	Decision metric
The harness beats a naive LLM ranking	`XP-01` LLM ranks raw quotes vs. harness	30 gold requests	Correct exclusions: 2/8 vs. 8/8
Two agents catch more than one	`XP-03` one combined agent vs. `AG-01` + `AG-02`	10 conflict cases	Conflicts detected at ≤ 1.5× tokens

10. Risks and open questions

- Both agents use the same LLM, so their blind spots may correlate → `XP-03`; consider a different model for `AG-02`.
- Weight profiles can be gamed → versioned, CPO sign-off, sensitivity shown in every memo.
- Who maintains `KG-01` ownership data after the pilot? (Decision asked of the CPO.)

UC2 Compare quotations like for like

1. Story: "As a **procurement analyst**, when **I receive 5 quotes in EUR, USD and CNY with different Incoterms**, I want **one landed-cost grid**, so that **I compare the true cost, not the headline price**."

2. User flow

Step	Human	AI	Tool / system	Autonomy
1	Selects the request	—	`TL-03` loads the quotes from `REPO-01`	Code
2	—	—	`TL-05`: price + freight + duty + FX (v3)	**Code**

Step	Human	AI	Tool / system	Autonomy
3	—	`MOD-02` adds the late-delivery probability	—	Suggests
4	**Reviews the grid; checks flagged lines**	—	`GR-01` marks expired quotes	Human

3. Context: `REPO-01` quotes + frozen FX table; `POL-01` for exclusions. **4. Tools:** `TL-03` (read), `TL-05` (compute).

5. Flow: `WF-01`, no LLM in the numbers. Using GenAI for arithmetic would be the wrong tool.

6. Guardrails: `GR-01` expired quotes are excluded; `GR-03` applies if a memo is later written.

7. Evals: `EV-01` landed cost = oracle on 80 quotes (100 %). **8. Observability:** FX version and quote validity per run.

9. Experiments: none: deterministic, so unit tests are enough.

10. Risks: missing freight terms → line flagged "incomplete", never assumed to be zero.

UC3 Related-party and compliance check

1. Story: "As a **compliance reviewer**, when **a supplier enters a shortlist**, I want **its ownership chain, certificate status and risk events, with sources**, so that **I catch related-party and expiry issues before the award, not after**."

2. User flow

Step	Human	AI	Tool / system	Autonomy
1	Opens the supplier	`AG-02` plans: owners → their risk events → certificates → clauses	`GR-05`	Acts alone (read)
2	—	—	`TL-02` up to 3 hops; `TL-04` restrictive clauses	Read
3	—	`MOD-01` summarises the findings (`SCH-02`, `GR-02`)	—	Suggests
4	**Reviewer accepts, rejects or requests verification**	—	Graph view highlights the path	Human

3. Context: `KG-01` (edges with provenance), `RAG-01` compliance statements, `PRM-02`. **4. Tools:** `TL-02`, `TL-04` (read).

5. Flow: agent `AG-02`: the path depends on what each hop reveals.

6. Guardrails: `GR-02` every claim cites an edge or a clause; `GR-06` on supplier statements; stale edges are shown as "unverified".

7. Evals: `EV-02` expected edges and provenance on 5 shared-parent + 5 expired-certificate traps (10/10); clause recall@5 ≥ 85 %.

8. Observability: `OB-01` traversal paths and the edges used.

9. Experiments: `XP-02` graph vs. text-only; prototype result 5/5 vs. 1/5 shared parents found.

10. Risks: ownership data from one registry only → add a second source; flag single-source edges.

UC4 Handle incomplete or conflicting evidence

1. Story: "As a **procurement analyst**, when **a quote says 30-day payment terms but the contract says 60**, or **a certificate is missing**, I want **the system to stop and show me the exact conflict**, so that **nobody approves on an invented or averaged fact**."

2. User flow

Step	Human	AI	Tool / system	Autonomy
1	—	`AG-01` / `AG-02` report findings with `conflict` set (`SCH-02`)	`TL-04` clause, `KG-01` edge	Acts alone (flag)
2	—	—	`GR-04` → status "needs review" in `MEM-01`	Code
3	**Analyst resolves: asks the supplier, picks the source, or excludes**	—	`OB-02` logs the resolution	Human

3. Context: `MEM-01` `unresolved_risks`; both sources linked (`RAG-01` clause, `KG-01` edge). **4. Tools:** `TL-04` (read).

5. Flow: the `WF-02` escalation branch. It is deterministic once a conflict is flagged.

6. Guardrails: `GR-04` never averages or picks a side; `GR-06` supplier claims cannot resolve their own conflict; a missing certificate = ineligible, never assumed (a rule in `POL-01`).

7. Evals: `EV-03` 5 contract conflicts + 5 missing certificates → all 10 flagged, 0 invented attestations.

8. Observability: count of conflicts per supplier, to find unreliable suppliers.

9. Experiments: none for V1. The rule is binary (flag or not); tune it only if false alarms exceed 20 %.

10. Risks: too many flags slow the process → measure the flag rate in the pilot; allow the manager to accept a documented exception.